

ABU SALEH MOSA FAISAL

Bioinformatics Analyst

Markey Cancer Center, University of Kentucky

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Education

College of Public Health, University of Kentucky

PhD in Epidemiology and Biostatistics

August 2025- Present

Lexington, Kentucky

Gatton College of Business and Economics, University of Kentucky

Master of Business Administration (MBA)

August 2023- May 2026

Lexington, Kentucky

New Mexico Institute of Mining and Technology

MS in Mathematics with specialization OR and Statistics

December 2021

Socorro, New Mexico

University Of Dhaka

BSc. in Applied Statistics

March 2013

Dhaka, Bangladesh

Academic Awards and Honors

- Awarded with scholarship to fund MS in Mathematics by New Mexico Institute of Mining and Technology, USA (2018).

Technical Skills

Programming Languages: R, Python, SAS, MATLAB, Java, C, C++

Databases : SQL, PostgreSQL, MySQL, NoSQL

Applications: Tableau, Power BI, IPA, MS Excel, Docker, GSEA, Minitab, L^AT_EX

Professional Experience

Markey Cancer Center, University of Kentucky

Bioinformatics Analyst, PI: Chi Wang, Ph.D.

March 2023 – Present

Lexington, Kentucky

- Deliver state-of-the-art biostatistics and bioinformatics support through advanced data analysis, study design, and interpretation of multi-omics datasets to assist basic science, translational, and clinical investigators.
- Collaborate with investigators to produce high-impact manuscripts and competitive grant proposals.
- Analyze diverse omics data (RNA-seq, ATAC-seq, ChIP-seq, scRNA-seq, scATAC-seq, etc.).
- Perform data mining using TCGA, cBioPortal, GDC, and Single Cell Portal.

LSU Health Science Center

Biostatistician, PI: Mohammad Alfred Nobel Bhuiyan, Ph. D.

May 2022 – March 2023

Shreveport, Louisiana

- Designed and executed statistical analyses for clinical trials and biomedical studies.
- Applied R, Python, and SAS to interpret complex datasets.
- Authored and reviewed statistical sections of manuscripts, abstracts, and posters.
- Provided consulting on study design, data analysis strategies, and interpretation.
- Led analysis of HCUP NIS hospital admission data to study substance use-related disease prevalence.

Research Experience

New Mexico Institute of Mining and Technology

Graduate Researcher, PI: Oleg Makhnin, Ph.D.

January 2018 – May 2021

Socorro, New Mexico

- Developed gibbSeq2, a Bayesian model for gene set enrichment analysis of RNA-seq data.
- Develop a Machine learning model to predict cancer treatment outcome
- Investigated serum creatinine trends in relation to phenacetin intake in women.

University of Dhaka

Research Assistant, Supervisor: M. Shafiqur Rahman, Ph. D.

Jan. 2013 – Dec. 2013

Dhaka, Bangladesh

- Analyzed factors contributing to low birth weight in Bangladeshi newborns.

Teaching Experience

New Mexico Institute of Mining and Technology

Graduate Teaching Assistant

August 2018 – December 2021

Socorro, New Mexico

- Instructed labs for Applied Statistics, Calculus I & II.
- Mentored undergraduate students and proctored examinations.

Publications

- Xing, Diensn G., Farhan Mohiuddin, Md Shenuarin Bhuiyan, Md Ismail Hossain, Zaki Al-Yafeai, Abu Saleh Mosa Faisal, Nicholas E. Goeders et al. "Prevalence and patterns of methamphetamine use and mental health disparity in the United States." *Nature Mental Health* (2024): 1-9.
- Khan, Muhammad Z., Rene Alvarez, Mohammad Alfrad Nobel Bhuiyan, Abu Saleh Mosa Faisal, Parker O'Neill, Muhammad Siddiqui, Praneet Kaki et al. "Clinical Outcomes of Transcatheter Aortic Valve Replacement (TAVR) vs Surgical Aortic Valve Replacement (SAVR) in Patients With Sarcoidosis." *Cureus* 16, no. 6 (2024).
- Rives, Taylor A., Abu Saleh Mosa Faisal, Jinpeng Liu, Hye Sook Chon, Casey Cosgrove, Bradley Corr, Robert Dood et al. "Comparison of the clinical and genomic profiles of endometrial cancer in Appalachian and non-Appalachian patients." (2024): 5591-5591.
- Pruitt, Aaron, Jinpeng Liu, Abu Saleh Mosa Faisal, Feitong Lei, Bin Huang, Eric B. Durbin, Jong Cheol Jeong et al. "Young lung adenocarcinoma: Molecular analysis pointing to early acquisition of aggressive mutations." (2024): e20089-e20089.
- Geisen, Mariah, Josiane weber-Tessmann, Courtney Kelson, Daheng He, Chi Wang, Abu Saleh Mosa Faisal, Jill Kolesar, and Yekaterina Zaytseva. "Inhibition of FASN postpones development of resistance to BRAF inhibitors in colorectal cancer." *Cancer Research* 84, no. 6, Supplement (2024): 4760-4760.
- Al-Yafeai, Zaki, Shafaqat Ali, Jimmy Brown, Maamannan Venkataraj, Md Shenuarin Bhuiyan, Abu Saleh Mosa Faisal, Kenneth Densmore et al. "Cardiomyopathy-associated hospital admissions among methamphetamine users: geographical and social disparities." *JACC: Advances* (2024): 100840.
- Mubashir, Maryam, Victoria Andrus, David Okuampa, Michelle Neice, Elizabeth Armstrong, Hailey Canezaro, Ross Dies et al. "Short-term Outcome of Peroral Endoscopic Myotomy Performed by the Same Endoscopist on Achalasia and Nonachalasia Esophageal Motility Disorders." *Surgical Laparoscopy Endoscopy & Percutaneous Techniques* 33, no. 6 (2023): 577-582.
- Xing, Diensn G., Teresa Horan, Md Shenuarin Bhuiyan, Abu Saleh Mosa Faisal, Kenneth Densmore, Kevin S. Murnane, Nicholas E. Goeders et al. "Social-geographic disparities in suicidal ideations among methamphetamine users in the USA." *Psychiatry research* 329 (2023): 115524.
- Raza, Syed Musa, Daniyal Raza, Elizabeth Armstrong, Victoria Andrus, Grace Ann McCurdy, Francesca Weis, Michelle Neice,..., Abu Saleh Mosa Faisal et al. "SHORT-TERM OUTCOMES OF GERD IN GASTROPARESIS PATIENTS AFTER GASTRIC PER ORAL ENDOSCOPIC PYLOROMYOTOMY-FINAL RESULTS." *Gastrointestinal Endoscopy* 97, no. 6 (2023): AB753.
- Khan, Jahidur Rahman, Md Mazharul Islam, Abu Saleh Mosa Faisal, Humayera Islam, and K. Shuvo Bakar. "Quantification of urbanization using night-time light intensity in relation to women's overnutrition in Bangladesh." *Journal of Urban Health* 100, no. 3 (2023): 562-571.
- Bhuiyan, Md Shenuarin, Abu Saleh Mosa Faisal, Maamannan Venkataraj, Nicholas E. Goeders, Steven R. Bailey, Steven A. Conrad, John A. Vanchiere, A. Wayne Orr, Christopher G. Kevil, and Mohammad Alfrad Nobel Bhuiyan. "Disparities in Prevalence and Trend of Methamphetamine-Associated Cardiomyopathy in the United States." *Journal of the American College of Cardiology* 81, no. 18 (2023): 1881-1883.
- Mubashir, Maryam, Syed Musa Raza, Shazia Rashid, David Okuampa, Daniyal Raza, Allison Derise, Kwabena O. Asafo-Agyei, Abu Saleh M. Faisal, Jehan Abdulsattar, and Qiang Cai. "Sa1473 POST-CHOLECYSTECTOMY SYNDROME; REVALUATING AND REDEFINING." *Gastroenterology* 164, no. 6 (2023): S-398.
- Hafiz, Nazar, Aditya Vyas, Omar Khan, Janak Bahirwani, Syed Musa Raza, Abu Saleh M. Faisal, Mohammad Alfrad Nobel Bhuiyan et al. "Su1647 ANALYSIS OF EACH COMPONENT OF GASTROPARESIS CARDINAL SYMPTOM INDEX (GCSI) IN PREDICTING THE RISK OF GASTROPARESIS IN PATIENT POPULATION WITHOUT THE DIAGNOSIS OF GASTROPARESIS." *Gastroenterology* 164, no. 6 (2023): S-635.

- Khan, Jahidur Rahman, Abu Saleh Mosa Faisal, Sukanta Das, and Nabil Awan. "Identifying overnutrition risk groups in Bangladeshi married women based on sociodemographic factors: A classification and regression tree model." *Obesity Medicine* 33 (2022): 100425.
- Faisal, Abu Saleh Mosa. *gibbseq2: A Bayesian Model to Detect Differential Expression of Gene Sets in RNA-seq Data*. New Mexico Institute of Mining and Technology, 2021.

Conference Presentation

- "Comparison of the Clinical and Genomic Profiles of Endometrial Cancer in Appalachian and Non-Appalachian Patients". 2024 ORIEN Scientific Summit. Tampa, FL, USA
- "Comparing relative abundance of microbial families during bio-remediation, using High throughput data and mixed models". Integrative Omics Conference-2019, Santa Fe, NM, USA
- "Profile analysis of microbial community using one-way anova model". NMT Student Research Symposium-2019, Socorro, NM, USA

Leadership & Service

- General Secretary of Bangladesh Student Association at University of Kentucky.
- Volunteer for Registration Responsibilities in "International Conference on Applied Statistics-2014" (ICAS-2014).
- Member of Tech Policy Group at New Mexico Tech

Workshops & Training

- NCGR 2022 virtual Bioinformatics Workshop on Differential gene Expression
- NCGR Fall 2021 Virtual Bioinformatics Workshop on Metagenomics and Metatranscriptomics